

• CSEA • IIT GUWAHATI OPTIMISATION TRACK

HACKATHON PROBLEM STATEMENT

POWER PLAY

T20 Cricket League Scheduling

★★★★☆ SENIOR / OPEN

36-48 HRS

NP-HARD

THE SCHEDULING CHALLENGE

Multi-objective optimization in a high-stakes franchise league.

The scheduling of a franchise T20 cricket league is a highly constrained, multi-objective optimization problem. Over a 56-day tournament, eight franchise teams play a total of 56 matches. The primary objective is to maximize overall league earnings (consisting of broadcaster advertising bids and alternate home venue attendance bonuses) while minimizing operational travel costs and player fatigue penalties.

A successful schedule must balance broadcaster preferences for premium matchups on weekends against geographic distances and stadium blackout dates. Additionally, operational constraints such as mandatory rest periods and balanced match densities prevent scheduling teams in rapid succession or idling them away from their home bases for too long.

8	56	56	13	14	8
FRANCHISES	MATCHES	DAYS	VENUES	MATCHES / TEAM	WEEKS

PROBLEM STATEMENT

FORMAL STATEMENT

PowerPlay — The IPL Scheduling Problem

Given a single problem instance containing eight teams, thirteen venues, and calibrated parameters for 2 revenue streams and 6 penalty functions, produce a **schedule** (an assignment of each of the 56 matches to a date and venue) that satisfies all hard constraints and maximizes **total league earnings** (revenue minus penalties).

EVALUATION & SCORING

Hackathon Scoring Overview

Scoring is based on: the **given test case** (40 pts), a **hidden test case** released in the final 2 hours (40 pts), a presentation **explaining how the team reached their solution** (20 pts), and a bonus for **parametric solver generalizability** (20 pts).

Match Definition

Every pair of the 8 teams plays each other *exactly twice*: once at each team's home venue. This yields $\binom{8}{2} \times 2 = 56$ matches. You must determine the specific day each match is played, and (for teams with alternate homes) at which venue it takes place.

Complexity

- ▶ **Large Search Space:** 56 matches mapped to 56 days represents an NP-hard allocation challenge.
- ▶ **Interdependent Choices:** A slot choice affects weekend revenue (R-1), rest-day validity (H-4), travel transitions (C-1), and rest-day spacing (C-5) simultaneously.
- ▶ **History Dependence:** Several penalties (C-3, C-4, C-5) depend on the sequential ordering of matches preceding the current one.

FRANCHISES & VENUES

The teams, home arenas, alternate venues, and their spatial layout across India.

CODE	FRANCHISE NAME	PRIMARY HOME GROUNDS	ALTERNATE VENUE
MI	Mumbai Indians	Wankhede Stadium	MCA Stadium (Pune)
CSK	Chennai Super Kings	M. A. Chidambaram	None
RCB	Royal Challengers Bengaluru	M. Chinnaswamy	None
DC	Delhi Capitals	Arun Jaitley Stadium	Shaheed VNS (Raipur)
KKR	Kolkata Knight Riders	Eden Gardens	None
SRH	Sunrisers Hyderabad	Rajiv Gandhi Int'l	ACA-VDCA (Vizag)
RR	Rajasthan Royals	Sawai Mansingh	Barsapara (Guwahati)
PBKS	Punjab Kings	IS Bindra Stadium	HPCA (Dharamsala)



Alternate Home Ground Rules

Five teams (MI, DC, SRH, RR, PBKS) have alternate homes, totaling 13 venues (8 primary + 5 alternate). Three teams (CSK, RCB, KKR) play all home games at their primary stadium.

Hosting a match at an alternate venue nets a revenue bonus (R-2), but is capped at **two home matches per team**. Travel to alternate stadiums has higher logistic expenses compared to primary ones, creating a trade-off between the R-2 bonus and C-1/C-2 travel costs.

CALENDAR, TRAVEL & CONSTRAINTS

How time moves forward, how teams travel, and the strict rules governing feasibility.

The 56-Day Schedule

The league runs for 56 consecutive days with exactly one match per day. Each day has a unique integer index $d \in \{1, 2, \dots, 56\}$. Days are labeled by their week index as **Day-Week**, where:

$$\text{Day} \in \{\text{Sun}, \dots, \text{Sat}\}, \quad \text{Week} \in \{1, \dots, 8\}$$

Team Movement Rules

A team is considered to be at the venue of its first match before the tournament starts. Thus, no travel or fatigue penalties accumulate before its first game. Following the first match, travel is assumed to occur immediately next day after the game. Moving to the same venue back-to-back incurs zero travel cost (but stay penalties continue to accrue).

H1 • PAIR BALANCE

Each pair plays exactly twice, once at each home ground

For every pair of teams $\{A, B\}$, there must be exactly one match hosted by A and exactly one match hosted by B . Total = 56 matches.

H2 • DAILY ALLOCATION

Exactly one match is played on each of the 56 days

Every day $d \in \{1, \dots, 56\}$ must contain exactly one match in the output schedule. No empty days, no double-headers.

H3 • PROGRESSIVE BALANCE

All teams stay within 4 matches of each other at all times

Let $m_T(d)$ be the number of matches team T has completed by the end of day d . For all days d : $\max_T m_T(d) - \min_T m_T(d) \leq 4$.

H4 • MANDATORY REST

No team plays on two consecutive days

If team T plays a match on day d , they cannot play a match on day $d + 1$. The minimum rest gap between matches is 2 days.

H5 • ALT VENUE LIMIT

At most 2 home matches at any alternate home ground

For the five teams with alternate venues, no more than 2 of their 7 home matches may be scheduled at their alternate venue.

H6 • BLACKOUT DATES

No match may be scheduled during venue blackout dates

The schedule must respect all input blackout constraints. No matches are permitted on restricted (venue, day) pairs.

OBJECTIVE FUNCTION & REVENUE

The mathematical formulation used to grade schedules, alongside revenue streams.

OBJECTIVE FUNCTION

$$\begin{aligned} \mathcal{J} = & \sum_m R(m) + \sum_m B_{\text{alt}}(m) \\ & - \sum_T \left[\text{Travel}(T) + \text{Distance}(T) + \text{Stay}(T) + \text{Density}(T) + \text{Gap}(T) \right] \\ & - \lambda_{\text{eq}} \cdot \text{Equity} \end{aligned}$$

Variables & Units: $\mathcal{T}$ = 8 teams, $\mathcal{V}$ = 13 venues, $\mathcal{M}$ = 56 matches. Distances are in **km** and financial terms (bids, travel costs, penalties) are in **Cr** (Crore INR). Each match m has day $d(m)$ and venue $v(m)$. x_T represents the chronologically sorted match sequence of team T .

R1

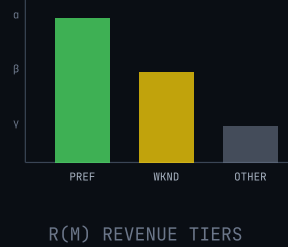
Broadcaster Revenue

THREE-TIER MATCHUP-SPECIFIC PAYOUT PER MATCH

For each scheduled match m with matchup $\{A, B\}$ (unordered pair of teams) on day $d(m)$:

$$R(m) = \begin{cases} \alpha_{\{A,B\}} & \text{if } d(m) \in \mathcal{P}_{\{A,B\}} \\ \beta_{\{A,B\}} & \text{else if } d(m) \in \mathcal{W} \\ \gamma_{\{A,B\}} & \text{otherwise} \end{cases}$$

Where $\alpha > \beta > \gamma \geq 0$ are the bids, $\mathcal{P}_{\{A,B\}}$ is the preferred days (with $\{A, B\}$ treated as an unordered pair), and $\mathcal{W}$ is the weekend day indices.



R2

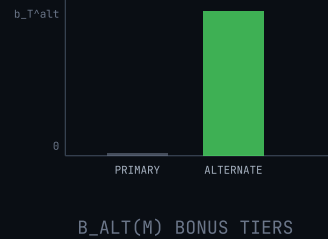
Alternate-Venue Bonus

ADDITIVE CASH BONUS FOR ALTERNATE STADIUM USAGE

Hosting a match m at home-team T 's alternate venue v_T^{alt} yields bonus b_T^{alt} :

$$B_{\text{alt}}(m) = \begin{cases} b_T^{\text{alt}} & \text{if } v(m) = v_T^{\text{alt}} \\ 0 & \text{otherwise} \end{cases}$$

Stacks additively with the broadcaster payout. This incentive balances the travel costs of scheduling away games.



TRAVEL & DISTANCE PENALTIES

Formulations targeting physical movement and late-tournament fatigue.

C1

Travel Cost

BASE CITY-TO-CITY COST SCALED BY LATE-TOURNAMENT MULTIPLIER

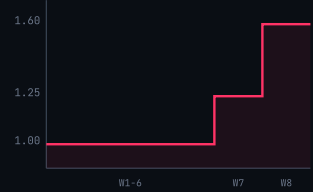
For team T with match venue sequence $v_1, \dots, v_{14}$:

$$\text{Travel}(T) = \sum_{k=1}^{13} 1[v_k \neq v_{k+1}] \cdot \text{cost}_{v_k, v_{k+1}} \cdot g(w_{k+1})$$

Where $\text{cost}_{u,v}$ is the base travel cost component from the travel matrix cell $M_{u,v} = [\text{dist}_{u,v}, \text{cost}_{u,v}]$, and g scales travel in late weeks to model postseason fatigue:

$$g(w) = \begin{cases} 1.00 & w \leq 6 \\ 1.25 & w = 7 \\ 1.60 & w = 8 \end{cases}$$

Key Strategy: Front-load expensive long-distance travels into weeks 1–6 to avoid the heavy 1.6x multiplier in the final week.



$G(W)$ MULTIPLIER STEPS

C2

Cumulative Distance Fatigue

CONVEX SCALING APPLIED TO TRAVEL DISTANCE EXCEEDING A THRESHOLD

Let $D(T)$ be team T 's total geographic distance traveled, calculated from the distance components of the travel matrix cells:

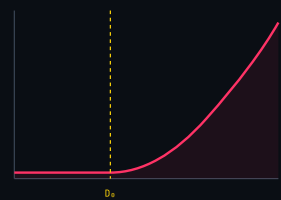
$$D(T) = \sum_{k=1}^{13} 1[v_k \neq v_{k+1}] \cdot \text{dist}_{v_k, v_{k+1}}$$

Where $\text{dist}_{u,v}$ is the distance component from the travel matrix cell $M_{u,v} = [\text{dist}_{u,v}, \text{cost}_{u,v}]$. The fatigue penalty applies convex scaling when $D(T)$ exceeds a baseline limit D_0 :

$$\text{Distance}(T) = \kappa \cdot [\max(0, D(T) - D_0)]^p$$

Where $p \in (1, 2]$ is the convexity exponent (typically $p = 1.5$), and $\kappa > 0$ is a scaling factor.

Key Strategy: Keeps teams from accumulating excessive total mileage. Short hops are preferred to keep $D(T)$ below or close to the baseline D_0 .



CONVEX FATIGUE ABOVE D_0

STAY & DENSITY PENALTIES

Formulations targeting hotel stays and match scheduling concentrations.

C3

Stay-at-Location Penalty

ESCALATING PENALTY FOR SPENDING CONSECUTIVE DAYS IN A NON-HOME CITY

For each stay e of length x_e days at non-home venue $v_e \neq H_T$:

$$\text{Stay}(T) = \sum_{\text{stays } e} 1[v_e \neq H_T] \cdot f_{\text{stay}}(x_e)$$

where $f_{\text{stay}}(x) = \eta \cdot \max(0, x - x_0)^q$

Where $x_0 = 2$ days, convexity $q \in [1.5, 2.5]$, and scaling $\eta > 0$. The penalty is zero if the stay occurs at the team's primary home ground H_T .

Key Trade-off: Balance the travel cost (C-1) of returning home against the stay penalty (C-3) of remaining in an away city during long match gaps.



STAY PENALTY EXCEEDING x_0

C4

Match-Density Penalty

STEPPED PENALTY BASED ON MATCH COUNT IN THE PRECEDING 7 DAYS

Let $\nu_T(m)$ be the count of team T 's matches scheduled within the 7 days prior to day $d(m)$:

$$\nu_T(m) = |\{m' \in \mathcal{M}_T : d(m) - 7 \leq d(m') < d(m)\}|$$

$$\in \{0, 1, 2, 3\}$$



STEPPED DENSITY PENALTY $\Delta(N)$

The penalty is calculated using matchup step costs:

$$\text{Density}(T) = \sum_{m \in \mathcal{M}_T} \delta(\nu_T(m)), \quad \text{where:}$$

$$\delta(\nu) = \begin{cases} \delta_0 & \nu = 0 \text{ (rust)} \\ 0 & \nu = 1 \text{ (optimal)} \\ \delta_2 & \nu = 2 \text{ (tired)} \\ \delta_3 & \nu = 3 \text{ (cooked)} \end{cases}$$

Key Strategy: Maintaining a density of $\nu = 1$ is ideal. The steep penalty δ_3 makes scheduling 3 matches in a 7-day window highly undesirable.

REST GAP & TRAVEL EQUITY

Formulations targeting rest consistency and travel balance across teams.

C5

Gap-Since-Last-Match Penalty

ASYMMETRIC PENALTY CENTERED AROUND OPTIMAL REST DURATION

For consecutive matches $m_T^{(k-1)}$ and $m_T^{(k)}$, let $\tau_k = d(m_T^{(k)}) - d(m_T^{(k-1)})$ be the gap in days. The penalty is formulated as:

$$\text{Gap}(T) = \sum_{k=2}^{14} \phi(\tau_k), \quad \text{where:}$$
$$\phi(\tau) = \begin{cases} a_{\text{low}} \cdot (\tau^* - \tau)^2 & \tau \leq \tau^* \\ a_{\text{high}} \cdot (\tau - \tau^*)^2 & \tau > \tau^* \end{cases}$$



ASYMMETRIC REST-GAP CURVE $\phi(T)$

Where optimal gap $\tau^* = 3$ days. The asymmetry coefficients satisfy $a_{\text{low}} > a_{\text{high}} > 0$ (typically $a_{\text{low}} = 3a_{\text{high}}$).

Key Strategy: A 3-day gap has zero penalty. Rest periods less than 3 days accumulate penalty costs rapidly, while larger gaps are penalized at a lower rate.

C6

Travel-Equity Penalty

LEAGUE-WIDE PENALTY TO PREVENT LOPSIDED TRAVEL ALLOCATIONS

Let $D_{\text{max}} = \max_T D(T)$ and $D_{\text{min}} = \min_T D(T)$ be the maximum and minimum total distances traveled across all teams. For threshold limit Δ_0 :

$$\text{Equity}(\text{schedule}) = \max(0, D_{\text{max}} - D_{\text{min}} - \Delta_0)$$



EQUITY PENALTY ABOVE LIMIT

This penalty scales with coefficient λ_{eq} in the master equation. It is the only global, league-wide penalty term.

Key Strategy: Prevents the optimizer from concentrating all travel on a single "sacrificial" team. Keeps the disparity within Δ_0 .

JSON DATA SCHEMAS & EVALUATION

The input/output file structures and the hackathon evaluation metrics. All distances are in **km** and finances are in **Cr** (Crore INR).

Input Files

Each evaluation instance consists of a directory containing the following five JSON files:

FILE NAME	DESCRIPTION & KEY SCHEMA COMPONENTS
teams.json	Definitions of the 8 franchises, their primary & alternate home grounds, and Alt Venue bonuses b_T^{alt} .
travel_matrix.json	A 13×13 symmetric matrix M where each cell contains a two-element array <code>[distance, cost]</code> representing city-to-city geographic distance (in km) and travel base price.
broadcaster_bids.json	Matchup valuations (α, β, γ) and preferred day sets $\mathcal{P}_{\{A,B\}}$ for all 28 pairings.
blackouts.json	An array of restricted <code>{"venue": "...", "day": d}</code> items where matches cannot be scheduled.
parameters.json	Coefficients $(\kappa, p, \eta, q, x_0, D_0, \delta_0, \delta_2, \delta_3, a_{\text{low}}, a_{\text{high}}, \tau^*, \Delta_0, \lambda_{\text{eq}})$ and multiplier vector g .

Output File Schema: `schedule.json`

Your solver must output a valid JSON containing `matches` (exactly 56 items):

FIELD PATH	TYPE	DESCRIPTION / VALUES
<code>matches[].day</code>	integer	Day index $d \in \{1, 2, \dots, 56\}$.
<code>matches[].home_team</code>	string	Designated home team code (e.g., <code>"MI"</code> , <code>"CSK"</code>).
<code>matches[].away_team</code>	string	Opposing team code.
<code>matches[].is_alt_venue</code>	boolean	True if the match is hosted at the alternate venue, false if hosted at the primary home venue.

Evaluation Metric & Scoring Rubric

Schedules are evaluated based on total league earnings $\mathcal{J}$. Stage scoring points will be calculated relative to the percentage of the best solution obtained overall among all teams.

EVALUATION COMPONENT	WEIGHT (PTS)	KEY CRITERIA
Given Test Case	40	Normalized score on the public dataset.
Hidden Test Case	40	Normalized score on the hidden dataset (released in the final 2 hours of the hackathon).
Solution Explanation	20	Presentation explaining the algorithms and core decision path.
Parametric Generalization (Bonus)	+20	Successful execution on perturbed, non-hardcoded datasets.

FAQ & GROUND RULES

Important questions answered, eligibility criteria, and code of conduct.

Frequently Asked Questions

Commercial Solvers? Yes, with a free community/academic license. Open-source solvers (OR-Tools, SCIP) are preferred.

Machine Learning? Allowed if models are trained within the build window or if the training pipeline is reproducible from the repo.

Hardware specs? Ubuntu box, 8 cores, 16 GB RAM, no GPU.

Hardcoding? Only team and venue codes. All parameters, coordinates, costs, bids, and blackouts must be read dynamically from inputs.

Ground Rules

- ❑ Teams must consist of 2 to 4 eligible students.
- ❑ All libraries, solvers, and utilities used must be open source.
- ❑ AI assistants (Copilot, Claude, Cursor) must be declared in the README.
- ❑ Pre-built dispatchers or copying prior work is strictly prohibited.

Build it like the BCCI deploys it.

The winning schedule is not the one with the most superficial features. It is the one that achieves optimal balance—maximizing broadcast payouts while minimizing travel expenses and fatigue under harsh physical limits.

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